# Estimating Cosmological Parameters from the Matter Power Spectrum using Amortized Bayesian Inference

— Project Report —
Simulation Based Inference

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# Contents

**1 Introduction** **1**

**2 Problem Definition** **1**

**3 Statistical model** **1**
3.1 Parameter Priors . . . . . 1
3.2 Likelihood . . . . . 2

**4 Data Generation** **2**

**5 Neural Posterior Estimation and Training** **2**

**6 Diagnostics** **4**
6.1 Parameter Recovery . . . . . 4
6.2 Simulation Based Calibration . . . . . 4
6.3 Posterior Contraction . . . . . 5
6.4 Other Configurations . . . . . 6

**7 Inference** **6**

**8 Discussion & Conclusion** **7**

**References** **8**

**Appendix: Additional Figures** **9**

# 1 Introduction

Good intro but the references are missing

Understanding the large-scale structure of the universe is central to modern cosmology, as it encodes important information about the composition, expansion history, and initial conditions of the universe. One of the most important quantities in this domain is the matter power spectrum, $P(k)$, which describes how matter density fluctuations vary across spatial scales $k$. Precise knowledge of the cosmological parameters that govern $P(k)$, such as the Hubble constant $(H_0)$, matter density $(\Omega_m)$, and spectral index $(n_s)$, is critical for constraining cosmological models like $\Lambda$CDM.

Traditional inference methods for these parameters often involve computationally expensive likelihood evaluations or simplifying assumptions, motivating the use of neural simulation-based inference approaches. This project applies BayesFlow, a likelihood-free neural posterior estimation framework, to recover the posterior distribution over $\theta = (H_0, \Omega_m, n_s)$ from noisy observations of the matter power spectrum. Using synthetic data generated from CAMB and adding noise, this report aims to demonstrate fast, accurate, and uncertainty-quantified parameter recovery. The approach offers an alternative to classical methods and supports ongoing efforts toward precision cosmology. The code is publicly available at https://github.com/smnehong/sbi_cosmological_parameters.

# 2 Problem Definition

The objective of this project is to estimate the cosmological parameters $H_0$, $\Omega_m$, and $n_s$ from noisy measurements of the matter power spectrum $P(k)$. The challenge lies in accurately inferring these parameters in the presence of observational uncertainty while avoiding the computational cost of traditional likelihood-based methods. To address this, the BayesFlow framework is employed to approximate the posterior distribution $p(\theta \mid y)$, where $y$ denotes the noisy observed $P(k)$, using simulated data generated with CAMB.

# 3 Statistical model

## 3.1 Parameter Priors

Good to use literature for priors but references are missing

The cosmological parameters were drawn from the following physically motivated priors:

$$
\begin{aligned}
H_0 &\sim \mathcal{N}(67.4,\, 10^2) \\
\Omega_m h^2 &\sim \mathcal{N}(0.1424,\, 0.035^2) \\
n_s &\sim \text{Gamma} \quad \text{such that } \mathbb{E}(n_s) = 0.965,\ \text{Var}(n_s) = 0.3^2
\end{aligned}
$$

These distributions ensure that a wide but physically plausible range of cosmologies is sampled during training. The specifics of the simulator require that $\Omega_m h^2$ is used instead of $\Omega_m$, which will be discussed in section 4. Because negative values for $n_s$ were not plausible in this context, the Gamma distribution was chosen to ensure positiveness.

### 3.2 Likelihood

To simulate observational uncertainty, Gaussian noise was added to the generated $P(k)$ with a standard deviation equal to a rough upper bound of $3\%$ of the maximum signal amplitude. Because we assume independent measurements at different wavenumbers $k$, we will sample the noise independently with the same $\sigma^2$ even though the different values for $P(k)$ are highly dependent on $k$. Writing $P(k)$ as a function of our parameters, the likelihood can be written as follows:

Good

$$y^{(k)} \mid \theta \overset{indep.}{\sim} \mathcal{N}(P_k(\theta), \sigma^2), \; k = k_1, \ldots, k_{200}$$

The biggest problem is that for a given $k$ we only have one observation for each $\theta$ without additional assumptions about $P_k(\theta)$. Therefore, we have to use the fact that the $y^{(k)}$ are not independent of each other and not exchangeable, when choosing a summary network for the approximator.

How did you choose the length of k?

# 4 Data Generation

Synthetic datasets of the matter power spectrum $P(k)$ were generated using the Python package `camb`, which implements the Code for Anisotropies in the Microwave Background (Lewis et al., 2000), a widely used code for calculating cosmological quantities such as the Cosmic Microwave Background (CMB) anisotropies and matter power spectra.

CAMB requires the parameters $H_0$, $n_s$, $\Omega_b h^2$, and $\Omega_c h^2$, where $\Omega_b$ and $\Omega_c$ denote the baryon and cold dark matter density fractions, respectively, such that $\Omega_b + \Omega_c = \Omega_m$. Because $h = H_0 \times 10^{-2}$, directly sampling $\Omega_m$ introduces a strong correlation between effective CAMB input parameters, which can hinder the learning of the inverse mapping. Initial experiments confirmed that training on such correlated inputs led to parameter recovery no better than random guessing for some cases.

To avoid this issue, $\Omega_m h^2$ was treated as an independent parameter. The baryon fraction

$$\frac{\Omega_b h^2}{\Omega_m h^2} \approx \frac{0.0224}{0.0224 + 0.12} \approx 0.1573$$

was held constant, and is consistent with the results of the Planck 2018 collaboration (Aghanim et al., 2020). Sampling from probability distributions was realized with `numpy` (Harris et al., 2020). Because $P(k)$ is of the order $10^5$, the logarithm is taken after adding the noise. Prior distributions and samples from the likelihood are shown in Figure 5 and Figure 6 in the appendix.

# 5 Neural Posterior Estimation and Training

BayesFlow, a Python framework for Amortized Bayesian Inference (ABI), introduced by Radev et al., 2023, was employed to train a neural network estimator of the posterior distribution

$$\hat{p}(\theta \mid h(y)) \approx p(\theta \mid y),$$

where $h(y)$ denotes a learned sufficient summary statistic. The training procedure consisted of two components: a summary network, which optimized the mapping $h(y)$, and an inference network, which approximated the posterior distribution. Amortization refers to the posterior not being approximated separately for each sample but learned as a function of the parameters, generally leading to slower training and faster inference.

The network architecture is summarized in Table 1. To capture autocorrelation within the data we used an LSTNet architecture as our summary network, implemented as `TimeSeriesNetwork` in BayesFlow which applies a one-dimensional Convolutional Neural Network to the data, after which either a bidirectional Long Short Term Memory (LSTM) or Recurrent Neural Network (RNN) architecture is used to capture temporal dependencies in the data. In this study, an LSTM network is used as the backbone model.

As an inference network, our model is using FlowMatching with a Multilayer Perceptron (MLP) subnet, which not only had the fastest training time, but also provided superior results to CouplingFlow. The model was trained using the configurations in Table 2. Because sampling from our simulator took around one second per sample, we decided for offline training, with a dataset of only 1024 samples and 64 validation samples. Disregarding the training data generation, the training itself also took more than 10 minutes per 100 epochs for FlowMatching and significantly longer for CouplingFlow, which is why configurations could only be compared qualitatively based on the criteria discussed in section 6. What is significantly longer?

**Table 1:** Model architecture used.

| Parameter | Value |
|---|---|
| Summary Network | LSTNet |
| CNN Kernel Size | 12 |
| Subnet | LSTM |
| Dropout | 0.05 |
| Activation | ReLU |
| Inference Network | FlowMatching |
| Subnet | MLP |
| Dimension | 256 |
| Layers | 7 |
| Dropout | 0.05 |
| Activation | ReLU |

**Table 2:** Training configuration for the model.

| Parameter | Value |
|---|---|
| Training samples | 1024 |
| Validation samples | 64 |
| Epochs | 100 |
| Optimizer | Adam |
| Standardization | all |

Figure 7 in the appendix shows the training loss dropping rapidly in the first few epochs, after which both training and validation curves gradually and smoothly stabilize, indicating convergence and good generalization by epoch 100. Because overfitting did not occur at any point in time, the default dropout rates of 0.05 were left unchanged. Early stopping was not used

# 6 Diagnostics

To assess the network we made use of parameter recovery plots, Simulation-Based Calibration (SBC), and posterior z-score contraction plots. These visualizations provide insight into predictive accuracy, calibration, and information gain across the inferred cosmological parameters.

## 6.1 Parameter Recovery

Figure 1 shows the model's estimated parameters plotted against ground truth values for $H_0$, $n_s$, $\Omega_m h^2$. Each subplot shows the true values on the x-axis, posterior medians on the y-axis, and a dashed diagonal for perfect agreement. Uncertainty, quantified by the MAD, is shown as a vertical line at each point. For $H_0$, estimates show strong correlation ($r = 0.926$), and for both $n_s$ and $\Omega_m h^2$ points cluster tightly along the diagonal ($r = 0.97$ and $0.981$), indicating high accuracy. Overall, the high correlations confirm strong generalization and reliable cosmological parameter estimation. Which network? What configuration?

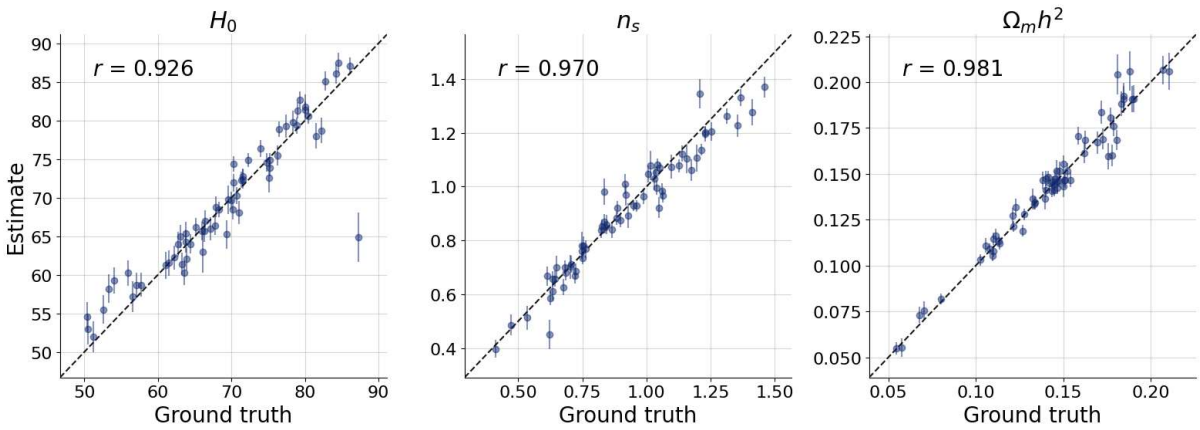


**Figure 1:** Ground truths vs. model predictions for $n = 64$ samples of unseen test data.

## 6.2 Simulation Based Calibration

SBC is a concept that uses the symmetry of $p(\theta \mid y)\, p(y) = p(\theta, y) = p(y \mid \theta)\, p(\theta)$ to reason that for $\theta^{(n)} \sim p(\theta)$ and $y^{(n)} \sim p(y \mid \theta^{(n)})$ a pair of parameter configuration and dataset is a sample of the joint distribution $(y^{(n)}, \theta^{(n)}) \sim p(y, \theta)$, implying the prior sample can be seen as one sample from the posterior given the generated data: $\theta^{(n)} \sim p(\theta \mid y^{(n)})$. If the approximator is able to sample from the true posterior, then for a number of samples $\hat{\theta}_j^{(n)} \sim \hat{p}(\theta \mid h(y^{(n)})) \approx p(\theta \mid y^{(n)})$, the prior sample $\theta^{(n)}$ that was used to generate the data should be uniformly distributed among the posterior samples $\hat{\theta}_j^{(n)}$. Figure 2 compares the empirical cumulative distribution function (ECDF) of the ranks with the uniform CDF. Across all parameters, deviations remain within the 95% confidence bands, giving no indication for miscalibrated estimates.

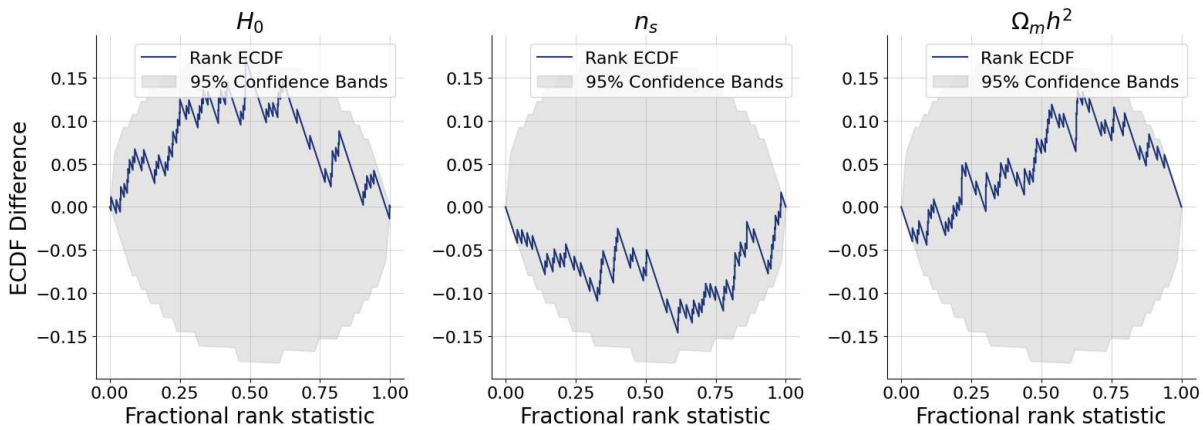


**Figure 2:** Difference between the rank ECDFs and the uniform CDF for $n = 64$ samples of unseen test data.

## 6.3 Posterior Contraction

Posterior contraction quantifies uncertainty reduction from prior to posterior, with values near 1 indicating high information gain, while the posterior z-score measures bias as the standardized offset between posterior mean and true value. Figure 3 shows that for all parameters, contraction values are close to 1.0, confirming strong uncertainty reduction. Z-scores mostly lie within ±2 and spread around 0, only $H_0$ shows an outlier at $z \approx -4$ which corresponds to the one in Figure 1, suggesting occasional sensitivity to specific extreme datasets. Overall, the model achieves informative, low-bias estimates, with rare extreme outliers warranting further review.

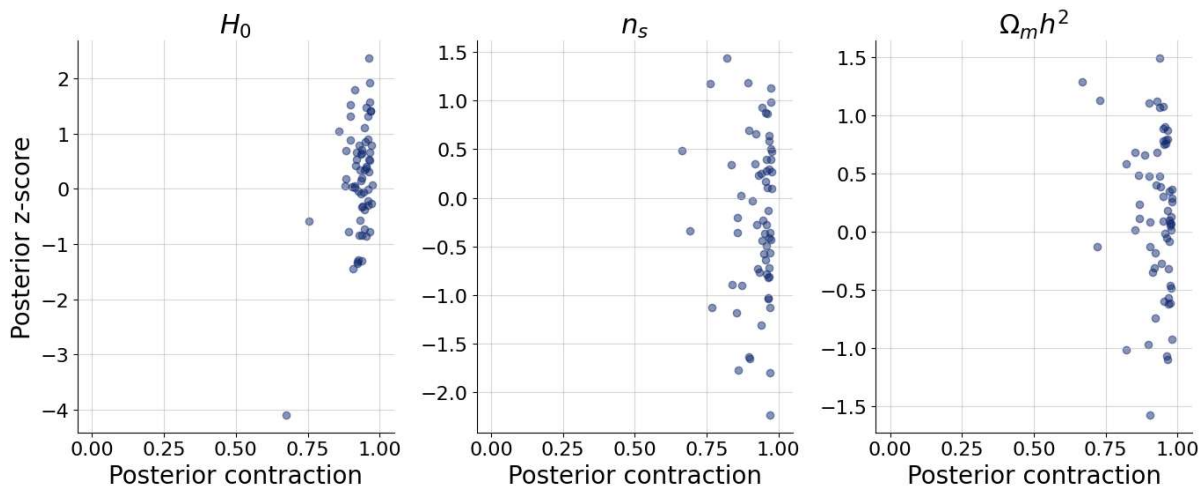


**Figure 3:** Posterior contraction and z-score for $n = 64$ samples of unseen test data.

### 6.4 Other Configurations

Diagnostic plots for notable other network architecures are included in the appendix. Here, SplineCoupling was chosen when CouplingFlow was used, because AffineCoupling was not able to learn the parameters sufficiently. With SplineCoupling, while parameter recovery is similar for the GRU and LSTM subnets in Figure 9 and Figure 10 respectively, the GRU version shows a significant bias towards lower parameter values for $H_0$ and $n_s$ when examining the RECDF difference plot in Figure 12 while SBC does not show miscalibration when using an LSTM subnet (see Figure 13).

Using FlowMatching instead of CouplingFlow with an LSTM improved fit for $H_0$ and $\Omega_m h^2$ (Figure 1 and Figure 10) and enhanced posterior contraction to a near optimal level (Figure 3 and Figure 16). Again, the LSTM subnet performed better than the GRU one. Combining FlowMatching with a GRU subnet led to the poorest performance of the four architectures in question, with low correlation of true values and predictions, and posterior contraction spread around 0.5 for all parameters. However, this performance is likely best attributed to the fact that for this specific configuration no standardization of the summary statistics was done and that a narrower prior for $n_s$ was used, limiting how different the data the model is trained on was.

## 7 Inference

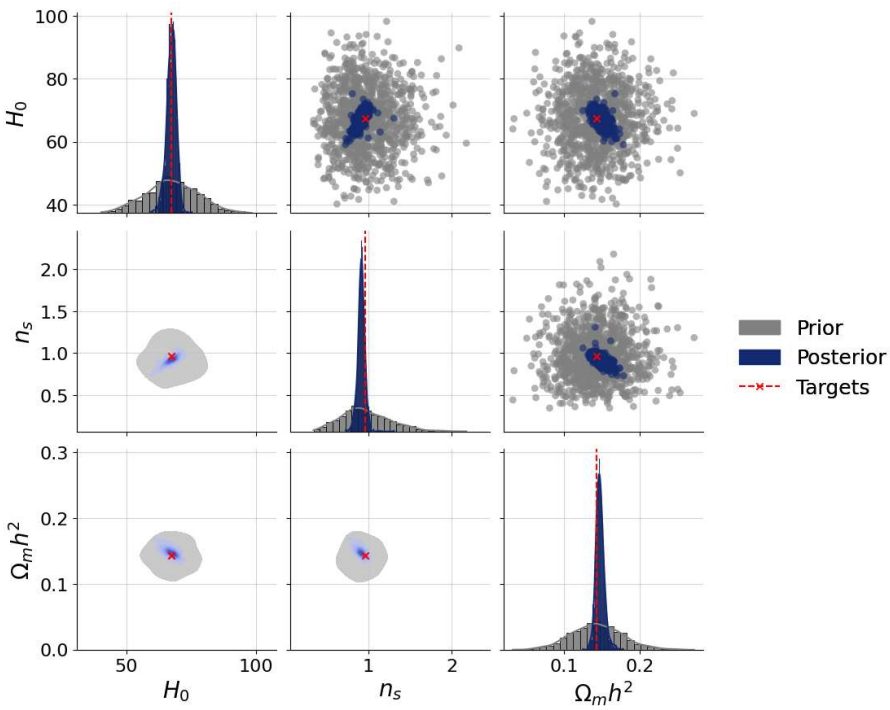


**Figure 4:** Posteriors and priors, with one likelihood sample generated from ground truth parameters by Aghanim et al., 2020.

For inference, we generated a single likelihood sample using the parameter estimates of Aghanim et al., 2020 as ground truth. Afterwards, 1024 samples were drawn from the posterior, given the likelihood sample. The resulting pairwise posterior plot compares the prior (gray) and posterior (blue) distributions for $H_0$, $n_s$, and $\Omega_m h^2$. In Figure 4, the sharper posterior peaks relative to the priors indicate contraction for this dataset, while the scatter plots show reduced spread and clustering around the true values (red), illustrating the model's ability to recover parameters which are close to the real universe.

# 8 Discussion & Conclusion

In conclusion, this project successfully applied ABI using BayesFlow to infer cosmological parameters from the matter power spectrum. As shown in section 6, posterior distributions over cosmological parameters $\Omega_m h^2$, $H_0$ and $n_s$ are approximated using a neural posterior estimator as described in section 5 using synthetic data. Parameter recovery is good for $H_0$ ($r = 0.926$) and excellent for $n_s$ ($r = 0.97$) and $\Omega_m h^2$ ($r = 0.981$). Simulation-based calibration does not indicate any miscalibration and the posterior consistently contracts well around the true parameter values.

This project was heavily limited by the slow data generation and training. Therefore, only a limited number of training configurations could be tested and compared qualitatively by evaluating SBC, fit and posterior contraction. Future work should therefore use a larger budget to quantitatively compare and optimize hyperparameters such as the parameters of the prior distributions, different model architectures and their respective hyperparameters.

We not only learned how to apply ABI to a real world problem but also had to deal with computational limitations. This led us to spend a lot of time trying to speed up computation and interpret the results. Specifically, we gained insight into the training and inference speed tradeoff between the FlowMatching and CouplingFlow architectures. In addition, the project helped us understand other key diagnostics beside simple parameter recovery and why they are needed.

- Good intro - defined the problem and motivated use of SBI
- Priors are physically plausible and physical constraints such as Baryon fraction was considered.
- Use the correct summary net - TimeSeries
- Use flow matching for inference net
- Offline training, no early stopping but loss curve looks good
- mention that FM perfroms better than CFs and provide the comparitive results
- Provide the necessary diagnostics and discuss them

1.0

# References

Lewis, A., Challinor, A., & Lasenby, A. (2000). Efficient computation of cosmic microwave background anisotropies in closed friedmann-robertson-walker models. *The Astrophysical Journal*, *538*(2), 473–476. https://doi.org/10.1086/309179

Aghanim, N., Akrami, Y., Ashdown, M., Aumont, J., Baccigalupi, C., Ballardini, M., Banday, A. J., Barreiro, R. B., Bartolo, N., Basak, S., Battye, R., Benabed, K., Bernard, J.-P., Bersanelli, M., Bielewicz, P., Bock, J. J., Bond, J. R., Borrill, J., Bouchet, F. R., … Zonca, A. (2020). Planck2018 results: Vi. cosmological parameters. *Astronomy & Astrophysics*, *641*, A6. https://doi.org/10.1051/0004-6361/201833910

Harris, C. R., Millman, K. J., van der Walt, S. J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M. H., Brett, M., Haldane, A., del Río, J. F., Wiebe, M., Peterson, P., … Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*(7825), 357–362. https://doi.org/10.1038/s41586-020-2649-2

Radev, Schmitt, Schumacher, Elsemüller, Pratz, Schälte, Köthe, & Bürkner. (2023). BayesFlow: Amortized Bayesian workflows with neural networks. *Journal of Open Source Software*, *8*(89), 5702. https://doi.org/10.21105/joss.05702

# Appendix: Additional Figures

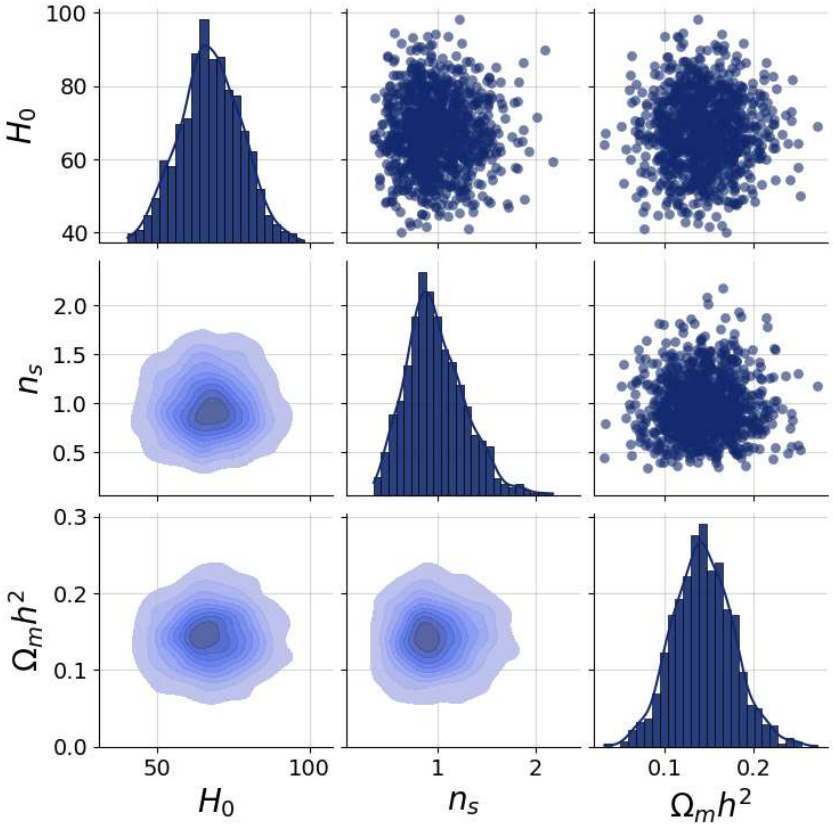


**Figure 5:** Marginal and pairwise joint distributions of the 1024 prior samples from the training dataset.

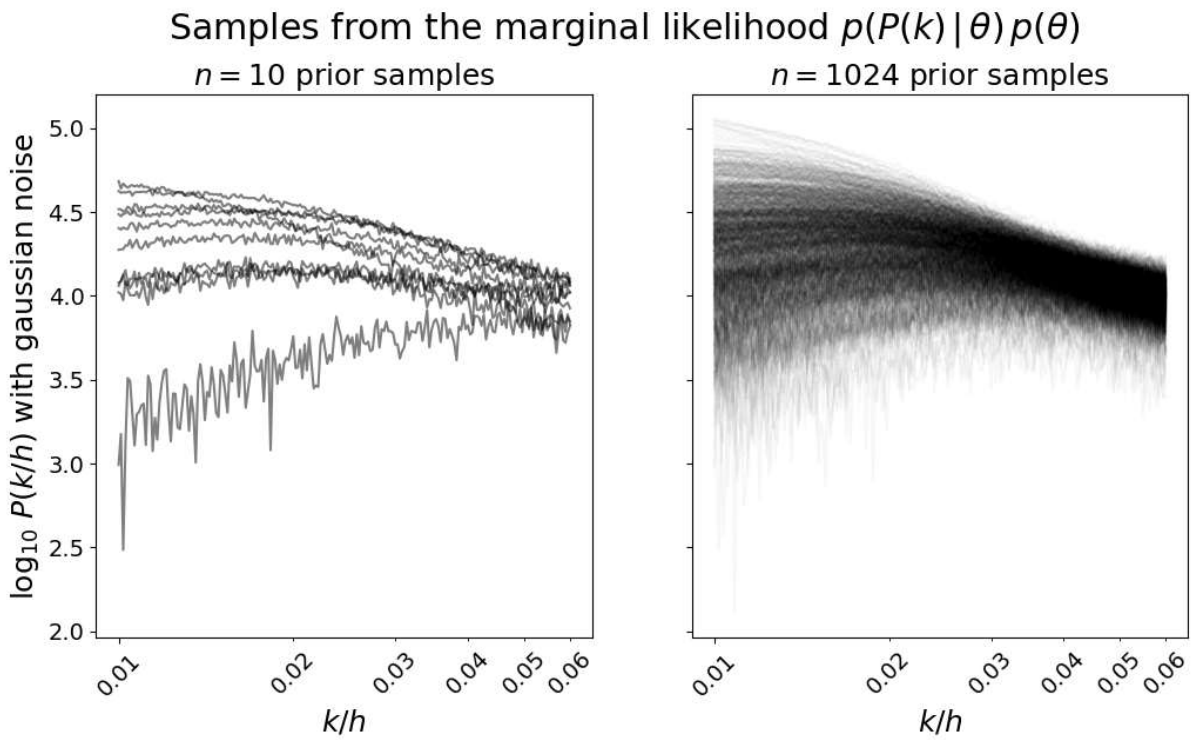


**Figure 6:** One likelihood sample each for ten prior samples (left) and for all 1024 prior samples from the training dataset (right).

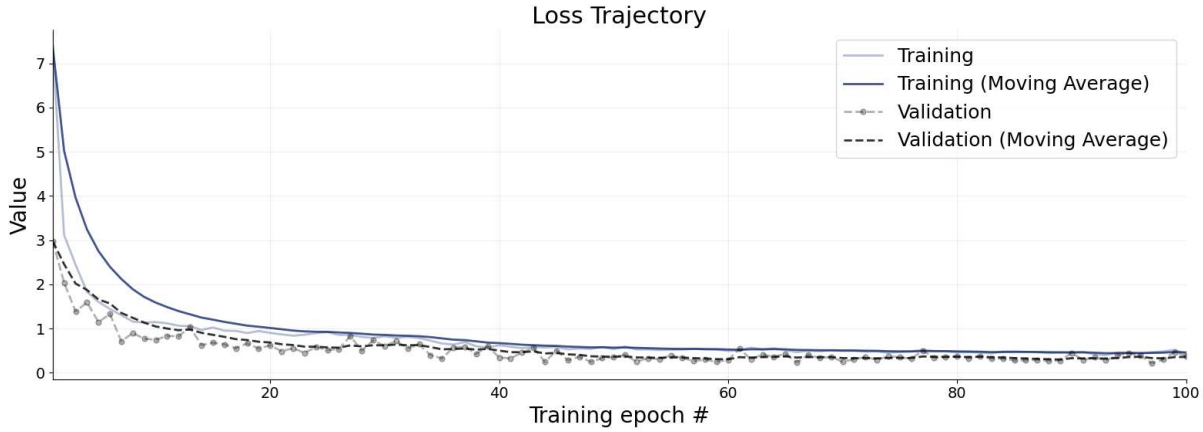


**Figure 7:** Loss on training and validation data of the LSTNet (LSTM) + FlowMatching model over 100 epochs.

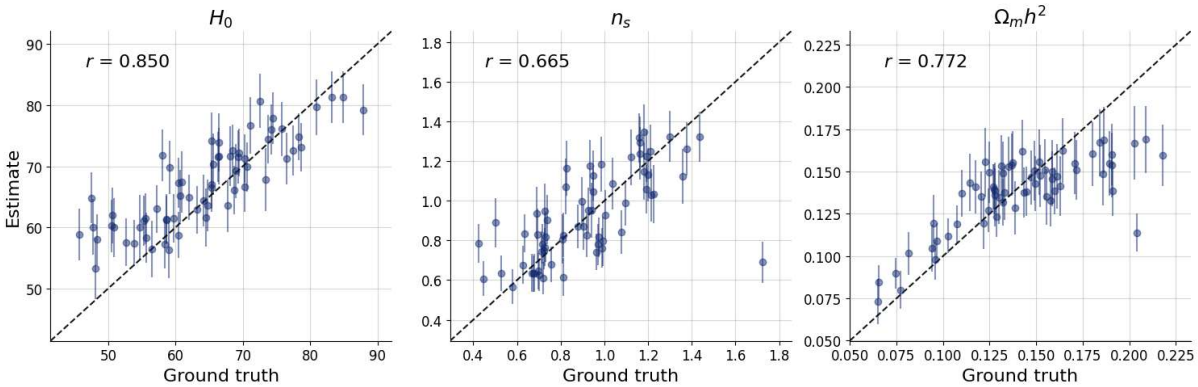


**Figure 8:** Parameter recovery of LSTNet (GRU) + FlowMatching trained for 200 epochs. In this configuration, only the inference variables were standardized and a standard deviation of 0.2 was used for the prior of $n_s$. Prediction errors for $\Omega_m h^2$ show a structure, with predictions reaching a plateau at 0.15.

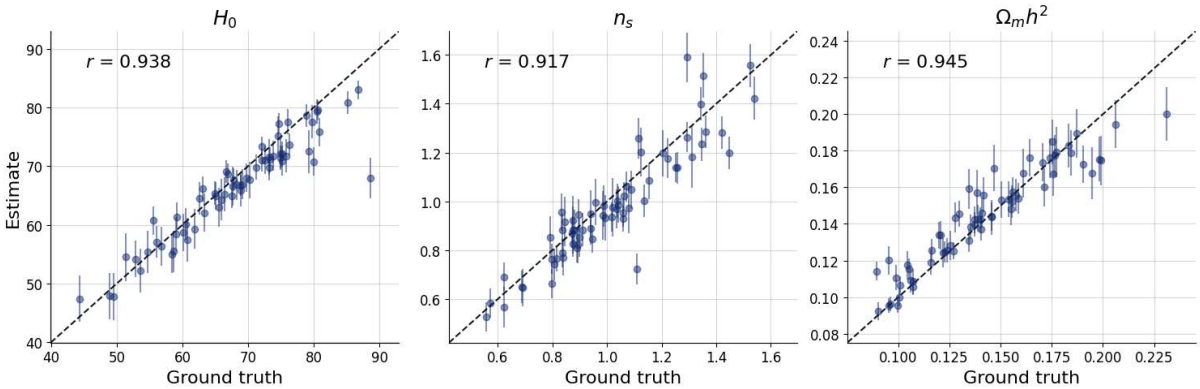


**Figure 9:** Parameter recovery of LSTNet (GRU) + CouplingFlow (SplineCoupling) trained for 100 epochs. $H_0$ and $n_s$ are consistently predicted lower compared to the ground truth.

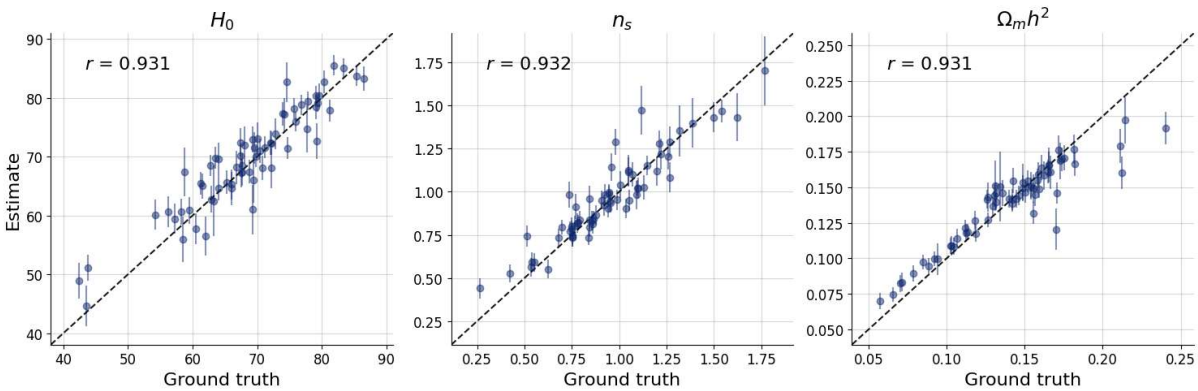


**Figure 10:** Parameter recovery of LSTNet (LSTM) + CouplingFlow(SplineCoupling) trained for 100 epochs.

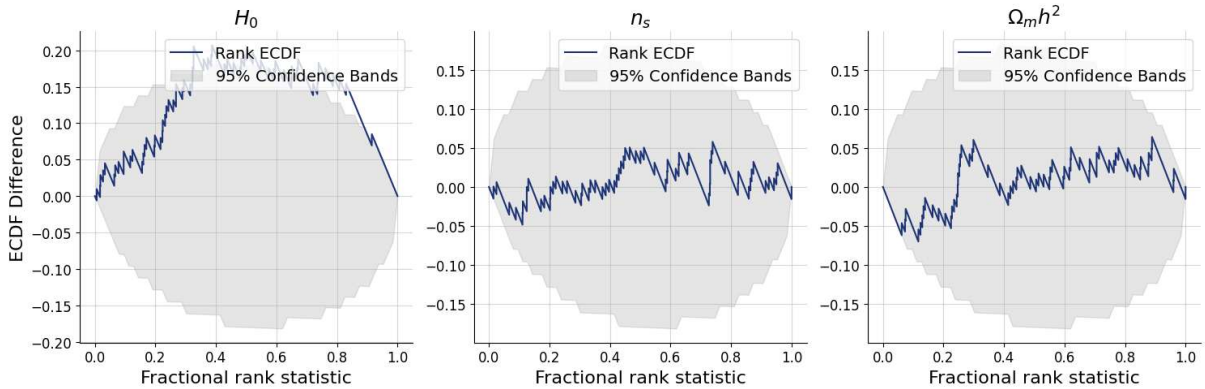


**Figure 11:** RECDF difference plot of LSTNet (GRU) + FlowMatching trained for 200 epochs. In this configuration only, just the inference variables were standardized and a standard deviation of 0.2 was used for the gamma prior of $n_s$.

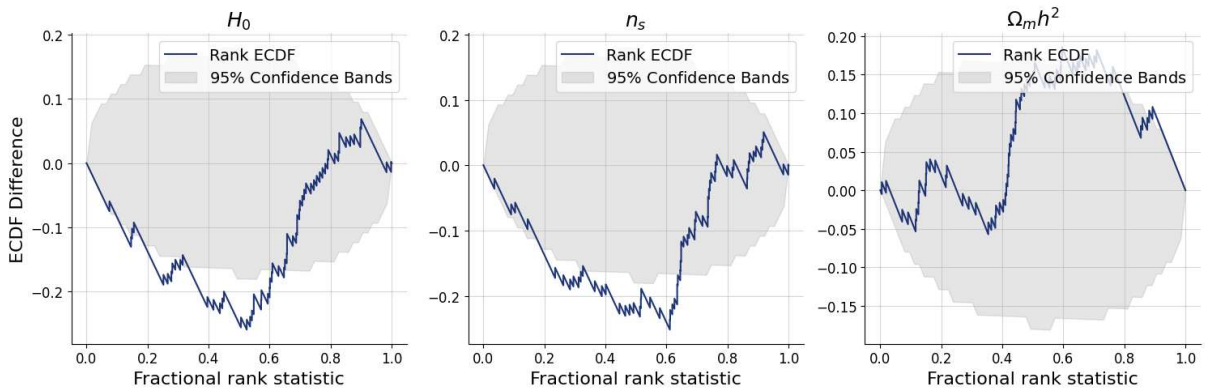


**Figure 12:** RECDF difference plot of LSTNet (GRU) + CouplingFlow (SplineCoupling) trained for 100 epochs.

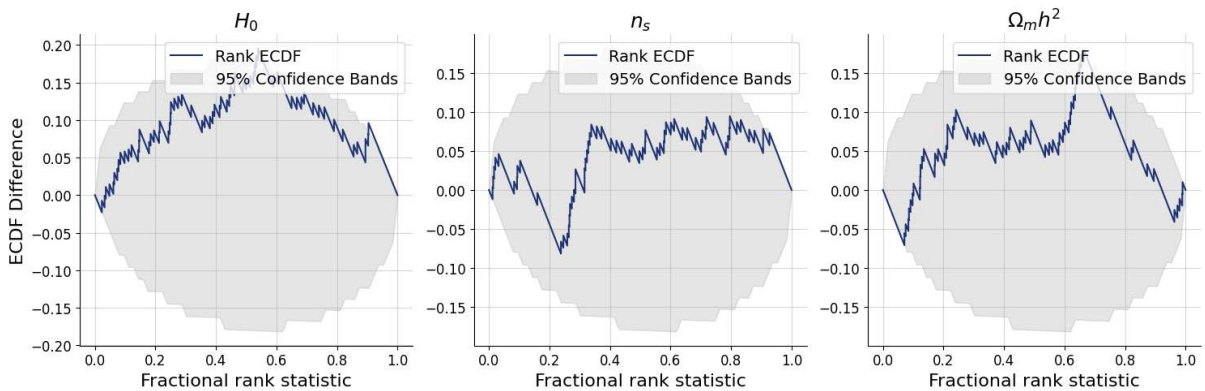


**Figure 13:** RECDF difference plot of LSTNet (LSTM) + CouplingFlow(SplineCoupling) trained for 100 epochs.

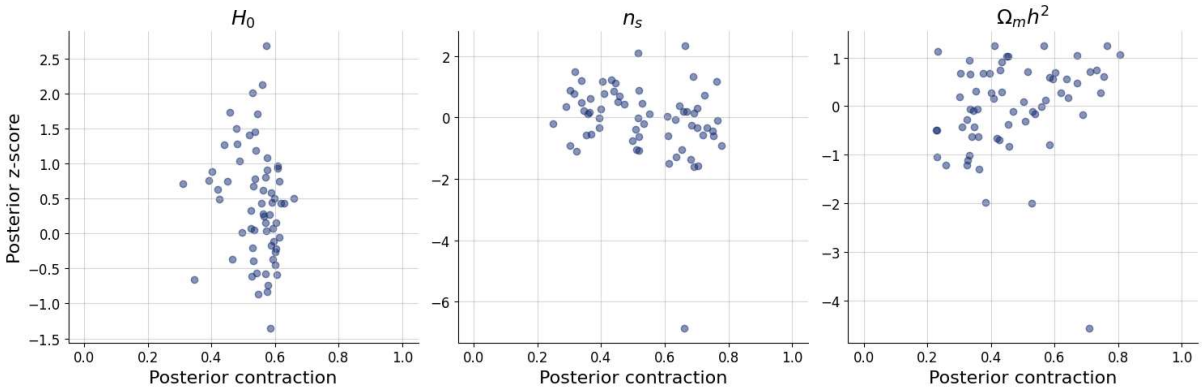


**Figure 14:** Posterior z-score against contraction of LSTNet (GRU) + FlowMatching trained for 200 epochs. In this configuration only, just the inference variables were standardized and a standard deviation of 0.2 was used for the gamma prior of $n_s$.

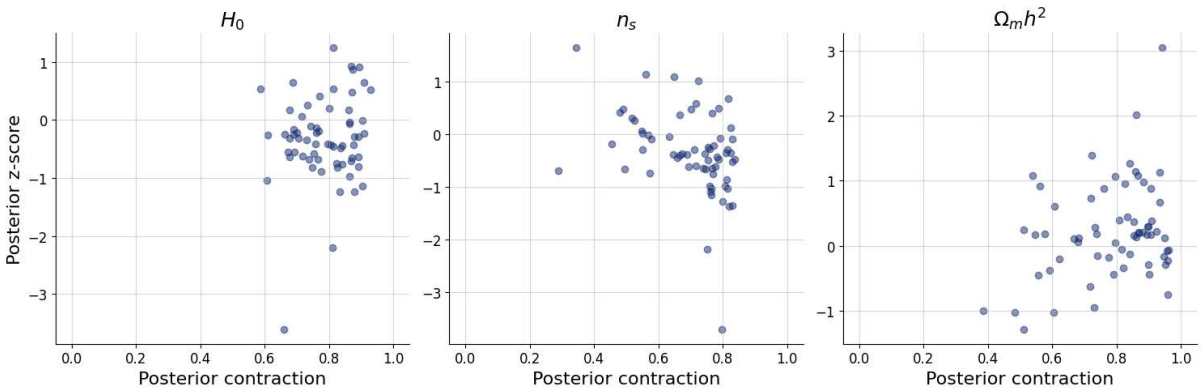


**Figure 15:** Posterior z-score against contraction of LSTNet (GRU) + CouplingFlow (SplineCoupling) trained for 100 epochs.

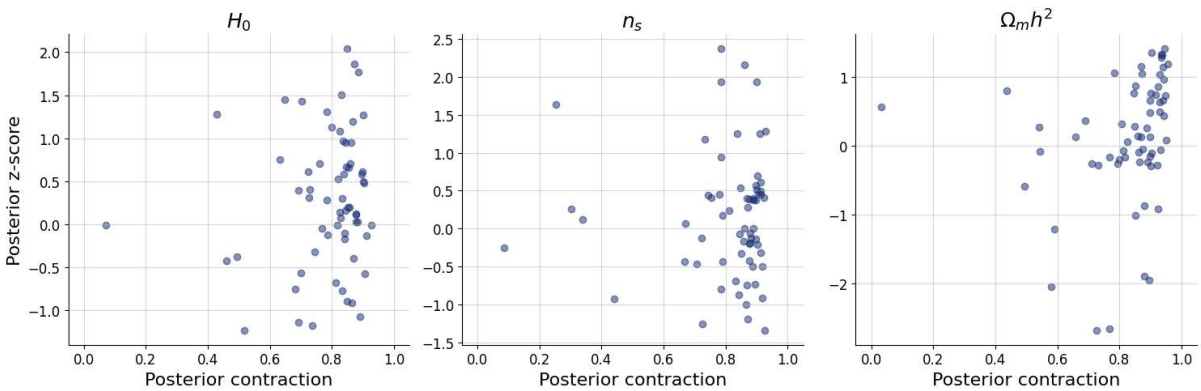


**Figure 16:** Posterior z-score against contraction of LSTNet (LSTM) + CouplingFlow(SplineCoupling) trained for 100 epochs.